2020

PAPER-III

GENERAL STUDIES AND CURRENT AFFAIRS

Time Allowed — 90 min. *Full Marks — 100*

If the questions attempted are in excess of the prescribed number, only the questions attempted first up to the prescribed number shall be valued and the remaining ones ignored.

1. Answer *any five* questions: 2×5=10

   (a) Write the laws of refraction of light.

   (b) Write with example the difference between mineral and ore.

   (c) Write two harmful effects of depletion of ozone layer on human health and environment.

   (d) What is hypermetropia? Which lens is used for its remedy?

   (e) What is meant by renewable energy source? Give an example.

   (f) Write down two uses of radioactivity.

   (g) Which type of electricity is used in electrolysis and why?

   (h) What are the bacteria called which decompose biomass to methane in biogas plant?

2. Answer *any five* questions: 2×5=10

   (a) Write two differences between hormone and nervous system regarding their mode of functions.

   (b) "Nitrogen Cycle is getting affected due to human activities".— State two phenomena to justify.

   (c) Write one cause and one symptom of lung cancer in human being.

   (d) Write two differences between plant and animal.

   (e) Write two functions of Mitochondria.

   (f) Give examples of two ways over population of human affects biodiversity.

   (g) How does reflex action help in daily life of human being?—Explain with two examples.

   (h) What is the relation between phenotype and genotype?

3. Answer *any ten* questions: 1×10=10

   (a) What is Metamorphic Rock?

   (b) What is the name of the steep ridge lying between two corries?

   (c) What is measured by Beaufort scale?

   (d) By which unit relative humidity is expressed?

   (e) Name any two gaseous pollutants.

   (f) What is the main reason for occurrence of tides?

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(g) Name the highest mountain peak of India.

(h) Name a cold desert region in India.

(i) Which state ranks first in per hectare production of rice in India?

(j) Name the petrochemical centre in West Bengal.

(k) Which is the longest national highway of India?

(l) Which is the sparsely populated state of India?

(m) Which is the most populous city of India?

4. Answer *any fifteen* questions: 2×15=30

(a) Define the term Nutrition.

(b) What do you mean by "malnutrition"?

(c) Define Health.

(d) What is RDA?

(e) What is the RDA for energy and protein for a coal mine worker?

(f) What is "Balanced Diet"?

(g) Define Energy.

(h) Write two units of energy.

(i) Mention two clinical symptoms of Vitamin A deficiency.

(j) Mention two clinical symptoms of iodine deficiency.

(k) Why sucrose is a non reducing sugar but maltose is not?

(l) What are monosaccharides?

(m) Write the differences between glucose and sucrose with clear structure.

(n) Define BMR.

(o) Name one aromatic amino acid and draw the structure.

(p) Write one important application of Iodine value.

(q) State any two functional roles of sugar in foods.

(r) State any two factors affecting protein bioavailability.

(s) State any two effects of too high intake of Proteins.

5. (a) Indian nationalism grew partly as a result of colonial policies and partly as a reaction to colonial policies. Discuss. 5

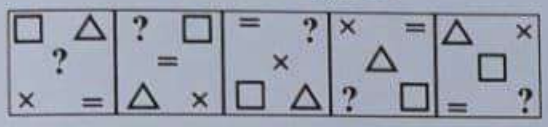
(b) "Though Swadeshi movement failed in achieving its immediate objective, yet it played a significant role in the Indian freedom struggle." Discuss. 5

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6. Answer *all* questions: 2×5=10

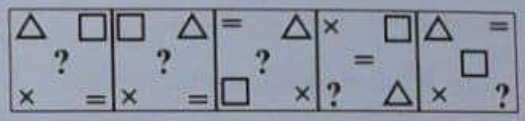
(a) Select a figure from amongst the Answer Figures which will continue the same series as established by the five Problem Figures.

Problem Figures



(A) (B) (C) (D) (E)

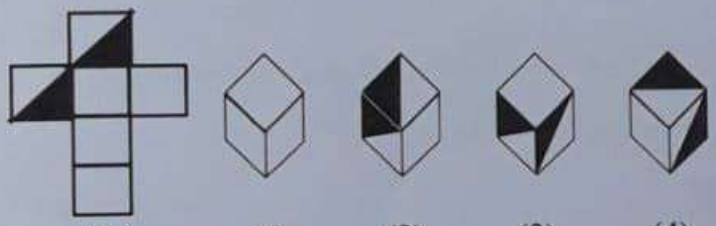
Answer Figures



(1) (2) (3) (4) (5)

Justify your answer.

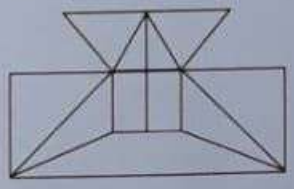
(b) Choose the box that is similar to the box formed from the given sheet of paper (X).



(X) (1) (2) (3) (4)

Justify your answer.

(c) Find the minimum number of straight lines required to make the given figure. Justify your answer.



(d) Look at this series : 8, 6, 9, 23, 87, ... What number should come next? Give reasons.

(e) Look at this series : 7, 10, 8, 11, 9, 12, ... What number should come next? Give reasons.

7. Answer *any five* questions: 2×5=10

(a) During the period of Covid outbreak, which state launched livelihood generation portal for returned migrant workers?

(b) Which fund is emphasized by National Education Policy 2020 to set up?

(c) Who was awarded the *2020 Nobel Prize in Physiology or Medicine*?

(d) What is the aim of PURVODAYA mission launched in West Bengal in January 2020?

(e) State the major research area of Nobel Laureate Dr. Abhijit Vinayak Banerjee in the field of Economics.

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(f) Mention any two objectives of 'Sabooj Sathi' project in West Bengal.

(g) Mention any two objectives of 'Karma Bhoomi' project in West Bengal.

(h) In which year Late Former President Pranab Mukherjee was awarded Bharat Ratna Award?

8. Answer *any five* questions: 2×5=10

(a) What are natural ecosystems?

(b) What is food web? Give example.

(c) What is meant by conservation of natural resources?

(d) What are fossil fuels?

(e) What is biodiversity?

(f) What are the main causes of the loss of biological diversity now-a-days?

(g) What are the air pollutants in automobile exhaust?

(h) What is essential in an EIA?

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PAPER-IV

ARITHMETIC

Time Allowed — 90 min. *Full Marks — 100*

If the questions attempted are in excess of the prescribed number, only the questions attempted first up to the prescribed number shall be valued and the remaining ones ignored.

Answer Question number 1 *and* any sixteen (16) *from the rest.*

1. Simplify: 4

$$\frac{4\sqrt{3}}{2-\sqrt{2}} - \frac{30}{4\sqrt{3}-\sqrt{18}} - \frac{\sqrt{18}}{3-2\sqrt{3}}$$

2. If $x = 1+\sqrt{2}+\sqrt{3}$ and $y = 1+\sqrt{2}-\sqrt{3}$, then find the value of $\frac{x^2+4xy+y^2}{x+y}$. 6

3. By selling 12 articles per rupee, there is a loss of 4%. How many articles should be sold per rupee so as to gain 44%? 6

4. The M.P. of a bicycle is ₹ 4000. While selling the bicycle, if three successive discounts of 20%, 10% and 5% are allowed, then what price will the customer pay for it? 6

5. Find the least square number which is exactly divisible by 3, 4, 5, 6, 8. 6

6. Difference of two perfect cubes is 189. If the cube root of the smaller number is 3, then find the cube root of the larger number. 6

7. Three bags contain 64·2 kg of sugar. The second bag contains $\frac{4}{5}$ of the contents of the first and the third contains $45\frac{1}{2}\%$ of what there is in the second bag. How much sugar is there in each bag? 6

8. If principal = ₹ 100000, rate of interest = 10% compounded half-yearly, find interest for 6 months and amount after 6 months. 6

9. During school hours, Neha finished 73% of her homework and Minakshi completed $\frac{5}{8}$ of her homework. Who must finish a greater per cent of homework? 6

10. Madhu's room measures 6m × 3m. Her carpet covers 8m^2. What per cent of floor is covered by the carpet? 6

11. If x varies inversely as y and $y = 60$ when $x = 1{\cdot}5$, find x when $y = 4{\cdot}5$. 6

12. If a deposit of ₹ 2000 earns an interest of ₹ 500 in 3 years, how much interest will a deposit of ₹ 36000 earn in 3 years with the same rate of simple interest? 6

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13. Shabnam takes 20 minutes to reach her school, if she goes at a speed of 6 km/h. If she wants to reach school in 24 minutes, what should be her speed? 6

14. A boy is cycling such that the wheels of the cycle are making 140 revolutions per hour. If the diameter of the wheel is 60 cm, then calculate the speed (in km/h) with which the boy is cycling. 6

15. Two cubes have volumes in the ratio 1 : 64. Find the ratio of the areas of a face of first cube to that of the other cube. 6

16. On dividing ₹ 200 between A and B, such that twice of A's share is less than 3 times B's share by ₹ 200, what is B's share? 6

17. The population of a town 2 years ago was 62500. Due to migration to cities, it decreases at 4% per annum. Find its present population. 6

18. Divide ₹ 2324 among three children in the ratio $1\frac{1}{4} : 1\frac{1}{3} : \frac{7}{8}$. 6

19. A, B and C working separately can do a work in 2, 3 and 4 days respectively. If they all work together and earn ₹ 3900 for the whole work, how should they divide the money? 6

20. Taps A and B can fill a tank in 4 hours and 6 hours respectively and tap C (at the bottom) can empty it in 12 hours. If all the three taps are opened together when the tank is empty, find after how many hours the tank will be full. 6

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